

Lets start off by a **numeric example** that we will approach its **solution** slowly, step-by-step. Later on, we will stretch our solution to dive deeper in the **theory** behind it in exactly **seven steps**.

Problem Statement

Given the data set below, figure out the which linear combinations matter the most out of these independent variables via Principle Component Analysis (PCA). Use PCA to reduce the given 2-dimensional data set.

Feature(s)	Example 1	Example 2	Example 3	Example 4
x	4	8	13	7
y	11	4	5	14

Figure (1)

Solution

Step 1: Determine no. of feature, no. of samples.

n: no. of features; $n = 2$

N: no. of samples; N = 4

Step 2: Computation of mean of variables.

❖ For variable x:

$$\bar{x} = \frac{4 + 8 + 13 + 7}{N} = \frac{4 + 8 + 13 + 7}{4} = 8$$

❖ For variable y:

$$\bar{y} = \frac{11 + 4 + 5 + 14}{N} = \frac{11 + 4 + 5 + 14}{4} = 8.5$$

Step 3: Computation of covariance matrix (for all ordered pairs)

- Ordered pairs (for variables x and y) are: (x,x) , (x,y) , (y,x) , (y,y)
 - Important Note: Having n features will give you n^2 pairs.
- Find covariance for each pair using the formula:

$$cov(x_i, x_j) = \frac{1}{N-1} \sum_{K=1}^N (x_{ik} - \bar{x}_i) (x_{jk} - \bar{x}_j)$$

- *Important Note: For a covariance matrix computed for the same variable:*

- $cov(x, x) = \frac{1}{N-1} \sum_{K=1}^N (x - \bar{x})^2$

- ❖ $cov(x, x) = \frac{1}{4-1} [(4-8)^2 + (8-8)^2 + (13-8)^2 + (7-8)^2] = 14$

- ❖ $cov(x, y) = \frac{1}{4-1} [(4-8)(11-8.5) + (8-8)(4-8.5) + (13-8)(5-8.5) + (7-8)(14-8.5)] = -11$

- ❖ $cov(y, x) = cov(x, y) = -11$

- ❖ $cov(y, y) = \frac{1}{4-1} [(11-8.5)^2 + (4-8.5)^2 + (5-8.5)^2 + (14-8.5)^2] = 23$

- Construct a (n x n) covariance matrix, S, for n = 2, covariance matrix will be (2 x 2):

$$S = \begin{bmatrix} cov(x, x) & cov(x, y) \\ cov(y, x) & cov(y, y) \end{bmatrix}$$

Which yields:

$$S = \begin{bmatrix} 14 & -11 \\ -11 & 23 \end{bmatrix}$$

Step 4: Computation of eigenvalues, eigenvectors, and normalized eigenvectors.

- I. Eigenvalues, λ (of a given covariance matrix):
 - $\lambda_1, \lambda_2, \dots, \lambda_m$, Where m is the number of newly reduced dimensions.
 - *Quick remarks on PCA in layman terms:*
 - PCA aims to fit straight lines to the data points.
 - We call these straight lines, “principle components”.
 - The 1st Principle Component (PC) is the best straight line you can fit to the data.
 - The 2nd PC, is the best straight line you can fit to the errors from the first principle component.
 - The 3rd PC is the best straight lines you can fit to the errors from the first and second principle components, and so on.
 - In Linear Algebra, a vector has a direction and a magnitude.
 - Some vectors will keep changing by a scalar value if you kept continuously multiplying it by a matrix of some sort, it might get longer or shorter and also rotate in either direction (clock-wise or counter clock-wise).
 - At some point in this multiplication process, vectors would only change their magnitudes (get longer or shorter), and will not rotate anymore. Vectors that has this property are called eigenvectors.
 - The scalar (scaling factor or magnitude of the vector; i.e.: when the vector gets longer or shorter) is called an eigenvalue.
 - Eigenvalue is the “variance” across a certain eigenvector.
 - Variance generally refers to how much data is “spread out” along a dimension or axis.
 - Eigenvectors and eigenvalues of a covariance (or correlation between two variables/features or more) matrix represent the “core” of a PCA. An eigenvector (principal component) determine the direction of the newly created (reduced) feature.

- An eigenvector (principle component) will keep turning until it finds the best linear combination (or recipe!) which fits best on the data, guaranteeing a maximal variance (spread) between these data point on that dimension.
- A principle component is quantified by an Eigenvector that points to the direction of the dimension with the greatest variance; hence PCA always tries to preserve as much variance as possible in the original data.

For a more illustrative animation, check the following GiF: (blue points are original data, red points are the projections of these data on the reduced feature/dimension which is the black line that keeps on rotating, which also represents the principle component)

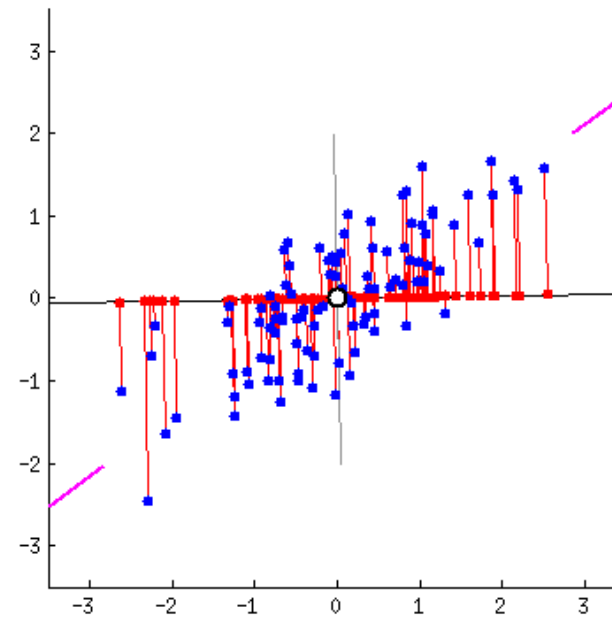


Figure (2)

- Using the covariance matrix determinant to get eigenvalues (λ):

$$\det(S - \lambda I) = 0$$

Equation 1

where:

S: covariance matrix

λ : eigenvalues

I: Identity matrix

To know more on how to calculate a matrix determinant:

Determinant of a Matrix

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- *Lets break down the terms from Equation 1:*

For the term I, since the covariance matrix is (2 x 2), we will need a (2 x 2) identity matrix as well:

*Let's break down the terms from **Equation 1**:*

For the term I, since the covariance matrix is (2 x 2), we will need a (2 x 2) identity matrix as well:

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

For the term λI :

$$\lambda I = \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$$

For the term $S - \lambda I$:

$$S - \lambda I = \begin{bmatrix} 14 - \lambda & -11 \\ -11 & 23 - \lambda \end{bmatrix}$$

Solve for λ :

$$\mathbf{det}(S - \lambda I) = 0$$

$$\det \left(\begin{bmatrix} 14 - \lambda & -11 \\ -11 & 23 - \lambda \end{bmatrix} \right) = 0$$

$$(14 - \lambda)(23 - \lambda) - (-11)(-11) = 0$$

$$322 - 14\lambda - 23\lambda + \lambda^2 - 121 = 0$$

$$\lambda^2 - 37\lambda + 201 = 0$$

Solve quadratic equation for roots:

$$\lambda = \{30.3849, 6.615\}$$

Sorting the roots in a descending, we have the following eigenvalues:

$$\lambda_1 = 30.3849, \lambda_2 = 6.615$$

II. Eigenvector e_i for each eigenvalue λ_i :

Solve for e_i , using the equation:

$$S \cdot e_i = \lambda_i \cdot e_i$$

For eigenvector e_1 which has the eigenvalue λ_1 :

- Important Note: to make e_1 compatible for multiplication with S, which is (2 x 2), we need to make sure that e_1 has 2 rows, hence it will be of dimensions (2 x 1), making the vertical vector composed of two components, $e_{1,1}$ and $e_{1,2}$, where the first sub-script stands for the first eigenvector, e_1 . These two components of the e_1 vector is referred to as the eigenvector's "linear combination".

$$S \cdot e_1 = \lambda_1 \cdot e_1$$

Sub for S and λ_1 :

$$\begin{bmatrix} 14 & -11 \\ -11 & 23 \end{bmatrix} \begin{bmatrix} e_{1,1} \\ e_{1,2} \end{bmatrix} = 30.3849 \begin{bmatrix} e_{1,1} \\ e_{1,2} \end{bmatrix}$$

We get two equations, one for the first row/component $e_{1,1}$ and one for the second row/component $e_{1,2}$:

$$14e_{1,1} - 11e_{1,2} = 30.3849 e_{1,1} \quad \text{Equation 2}$$

$$-11e_{1,1} + 23e_{1,2} = 30.3849 e_{1,2} \quad \text{Equation 3}$$

You can solve for $e_{1,1}$ and $e_{1,2}$ from any equation of them; you should get the same resultant eigenvector e_1 .

From Equation 2:

$$-11e_{1,2} = 30.3849 e_{1,1} - 14e_{1,1}$$

$$-11e_{1,2} = 16.385e_{1,1}$$

$$e_{1,2} = -1.49e_{1,1}$$

Sub $e_{1,1}$ for 1:

$$e_1 = \begin{bmatrix} 1 \\ -1.49 \end{bmatrix}$$

- Note: Any multiple of this vector will work. (An infinite number can satisfy this constraint)

III. Normalizing eigenvectors (Making it unit length)

The linear combination of e_1 :

$$e_1 = \begin{bmatrix} 1 \\ -1.49 \end{bmatrix}$$

- Quick remark:

- You can think of e_1 linear combination as a the recipe or the ratio between x and y features.
- In other words, for every 1 units we go along the x-feature axis, we go down 1.49 units along the y-feature axis.
- This means that data is mostly spread out along the y-feature axis.

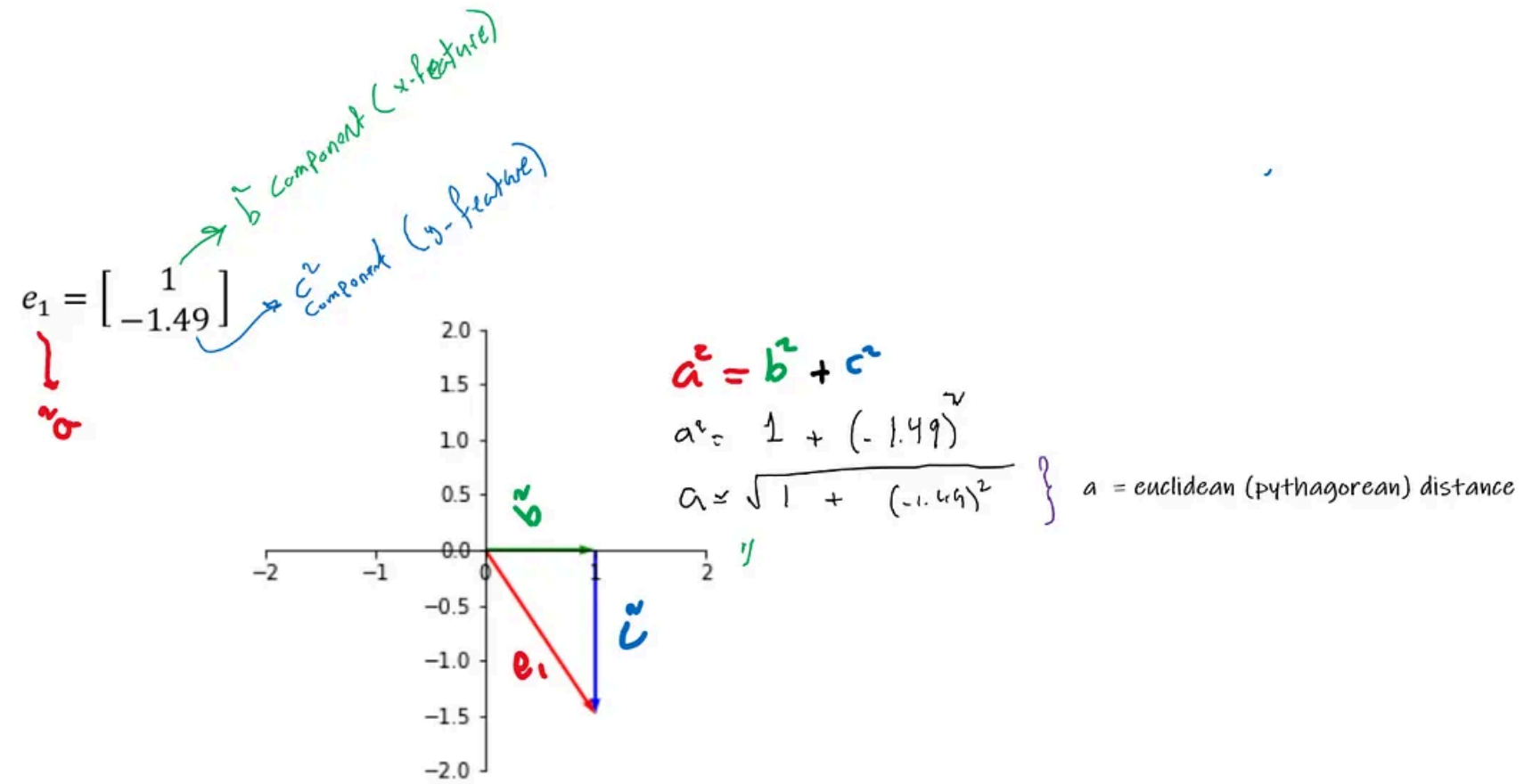


Figure (3)

- We solve for the length of the red line using the Pythagorean Theorem.

$$a = \sqrt{1^2 + (-1.49)^2} = 1.79$$

- When we do PCA, the recipe for PC1 is scaled so that the length of the red line = 1
- We can do that by dividing both components by the Euclidean distance.

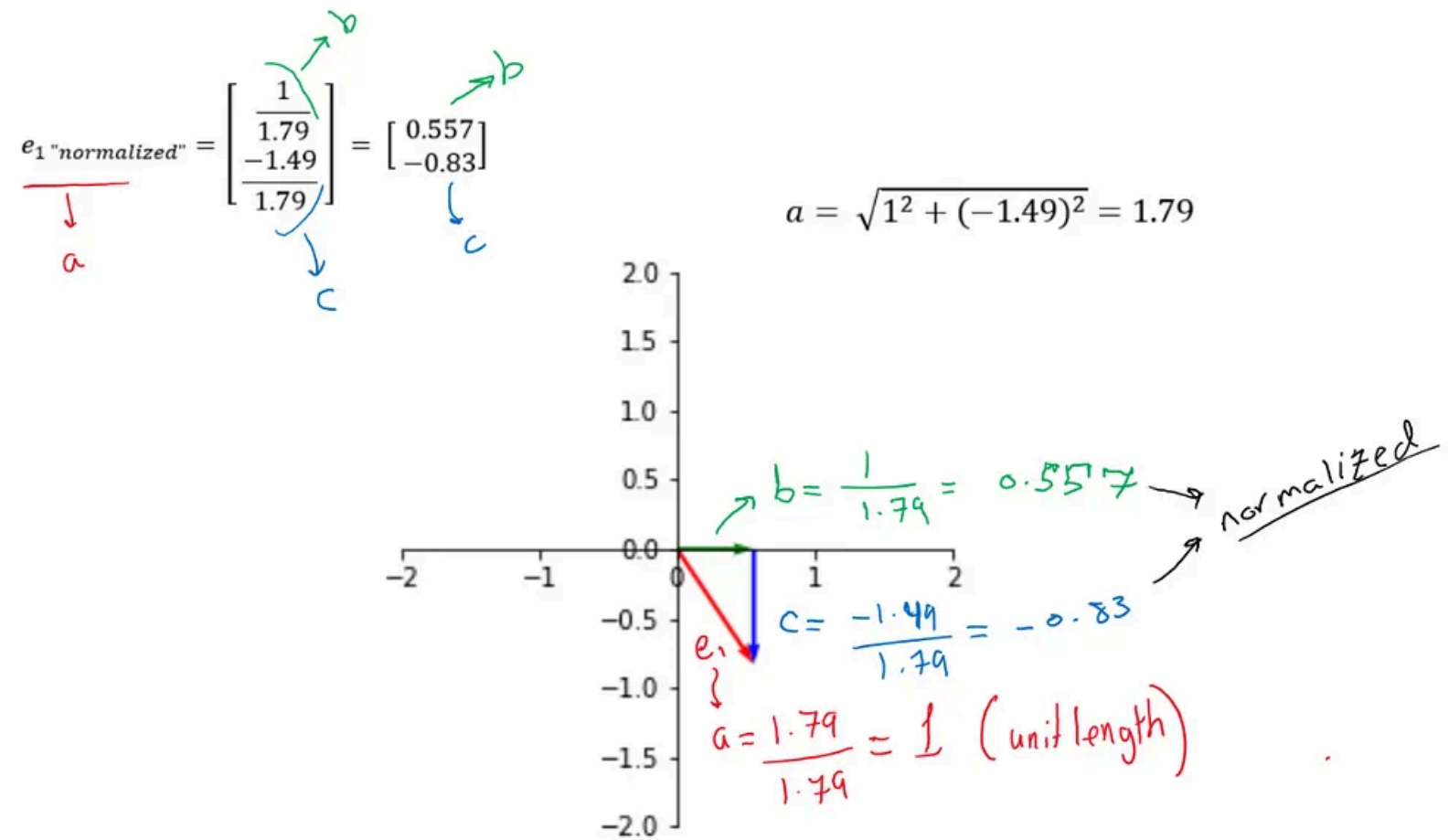


Figure (4)

Finally, we have the normalized e_1 of eigenvalue $\lambda_1 = 30.3849$ to be:

$$e_1 \text{ "normalized"} = \begin{bmatrix} \frac{1}{1.79} \\ \frac{-1.49}{1.79} \end{bmatrix} = \begin{bmatrix} 0.557 \\ -0.83 \end{bmatrix}$$

Instead of repeating the same process for e_2 , we can flip both components of e_1 and stick a minus sign in front of one of them.

So, normalized e_2 of eigenvalue $\lambda_2 = 6.615$ to be:

$$e_2 \text{ "normalized"} = \begin{bmatrix} 0.83 \\ 0.557 \end{bmatrix}$$

Step 5: How many dimensions (principle components)?

Now that we have e_1 and e_2 , we essentially have two principle components, PC1 and PC2 respectively. But should we use both to fit (project) our data?

ANSWER:

- In PCA, we try to preserve as much variance as possible, we do that by picking eigenvectors with the largest eigenvalues.
- To do that, we need to set a desired arbitrary threshold; let's say we want to capture 80% and above of the variance in our data; based on that we can choose to accept the component principles that explains proportions of variance that is higher or equal to that threshold, indicating which linear-combination (eigenvector) of the independent variables/features matter the most.
- Typical threshold values we can choose from: 80% $\rightarrow$ 95%

Compute percentage of variance each principle component is capturing from our data:

- ❖ For PC1: $\frac{\lambda_1}{\lambda_1 + \lambda_2} = \frac{30.3849}{30.3849 + 6.615} = 0.82$
 - PC1 “explains” 82% ($> 80\%$) of the overall variability.
- ❖ For PC2: $\frac{\lambda_2}{\lambda_1 + \lambda_2} = \frac{6.615}{30.3849 + 6.615} = 0.18$
 - PC2 “explains” 18% ($< 80\%$) of the overall variability.

Thus, we can pick PC1 to be the new reduced dimension, which points to the highest variance in our data, which in turn will be used to project our data on.

Step 6: Derive new data set

	Example 1	Example 2	Example 3	Example 4
PC1	P1,1	P1,2	P1,3	P1,4

Figure (5)

Compute each projection on PC1 from original data set:

- We can follow this formula to: 1) “center” (normalize) the data by subtracting the mean, and 2) “project” them by multiplying with the transpose of the normalized eigenvector:

$$e_j^T \cdot (x' - \bar{x}) \qquad \text{for } j = 1, \dots, m$$

where:

x' : sample from original data set

$\bar{x}$: mean of feature/variable

e_1^T : trnaspose of normalized eigenvector

m: number of reduced dimensions

$$P_{1,1} = e_1^T \cdot (x' - \bar{x})$$

❖ $P_{1,1} = [0.557 \quad -0.83] \begin{bmatrix} 4 - 8 \\ 11 - 8.5 \end{bmatrix} = -4.303$

❖ $P_{1,2} = [0.557 \quad -0.83] \begin{bmatrix} 8 - 8 \\ 4 - 8.5 \end{bmatrix} = 3.735$

❖ $P_{1,3} = [0.557 \quad -0.83] \begin{bmatrix} 13 - 8 \\ 5 - 8.5 \end{bmatrix} = 5.69$

❖ $P_{1,4} = [0.557 \quad -0.83] \begin{bmatrix} 7 - 8 \\ 14 - 8.5 \end{bmatrix} = -5.122$

- Populate new data set:

	Example 1	Example 2	Example 3	Example 4
PC1	-4.303	3.735	5.69	-5.122

Figure (6)

Step 7: Project to new dimension(s)

- Start with a default figure.
- Label the means of each feature. ($\bar{x} = 8$, $\bar{y} = 8.5$)

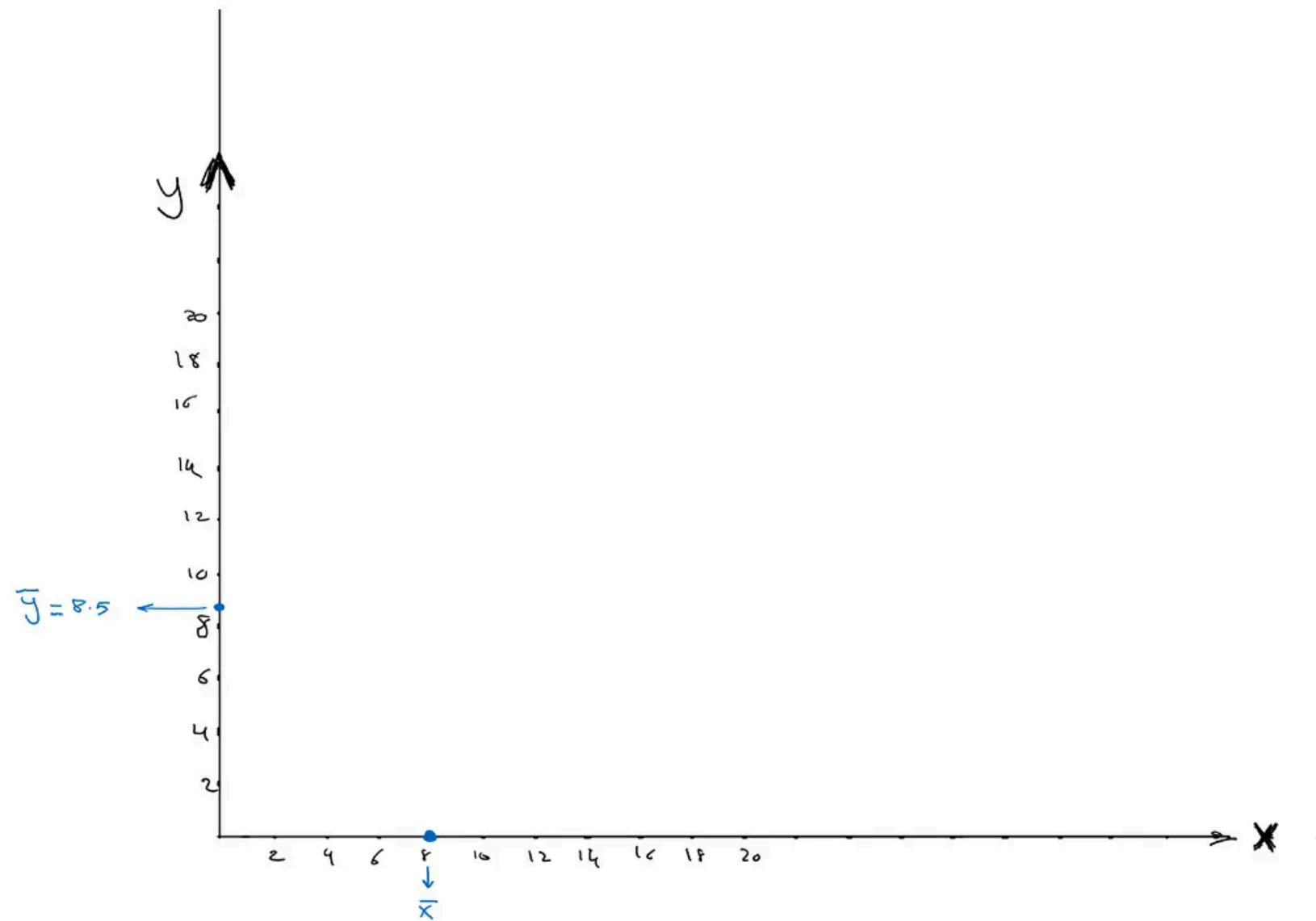


Figure (7)

- Normalize the coordinates to make sure data is centered. (i.e.: subtracting the mean)

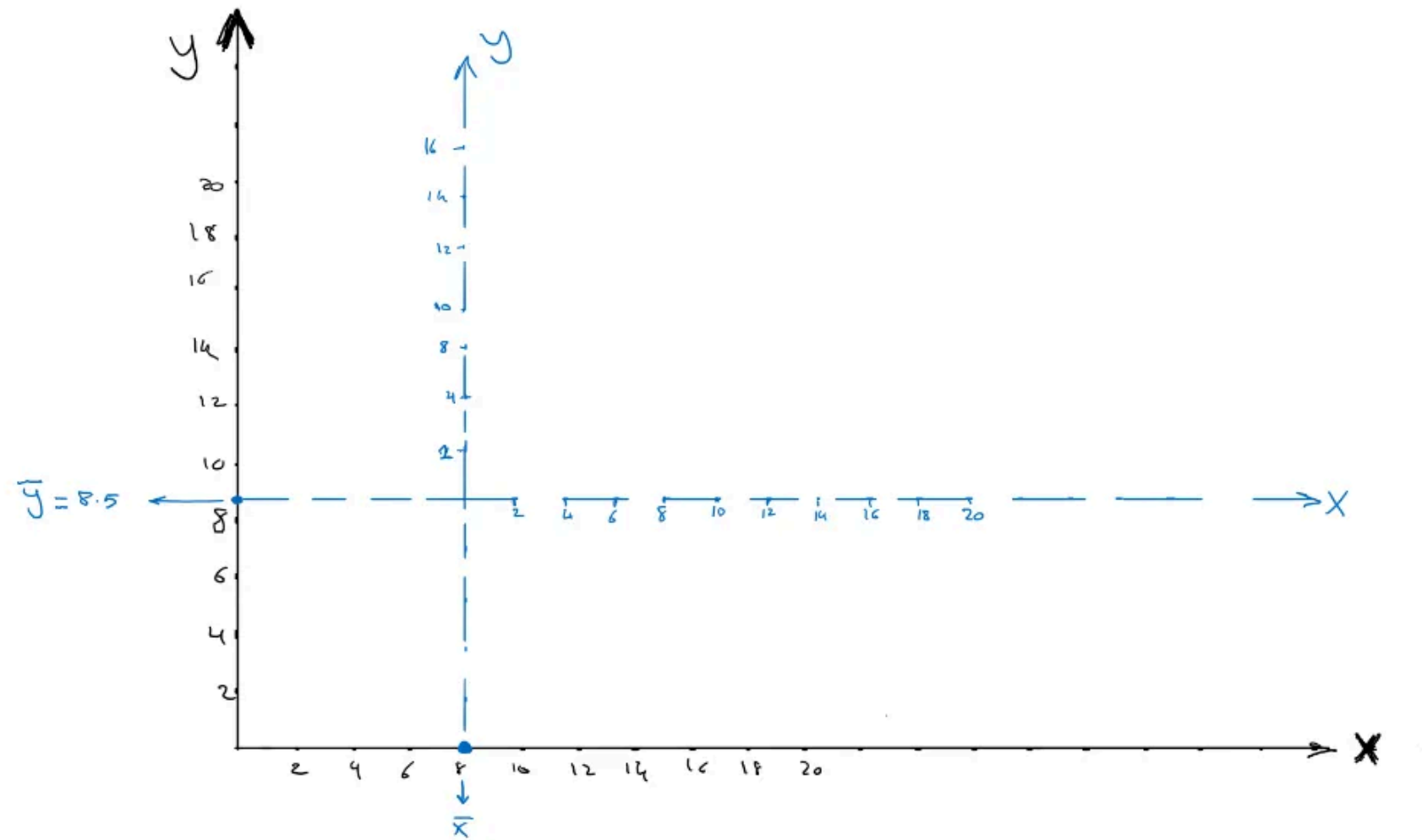
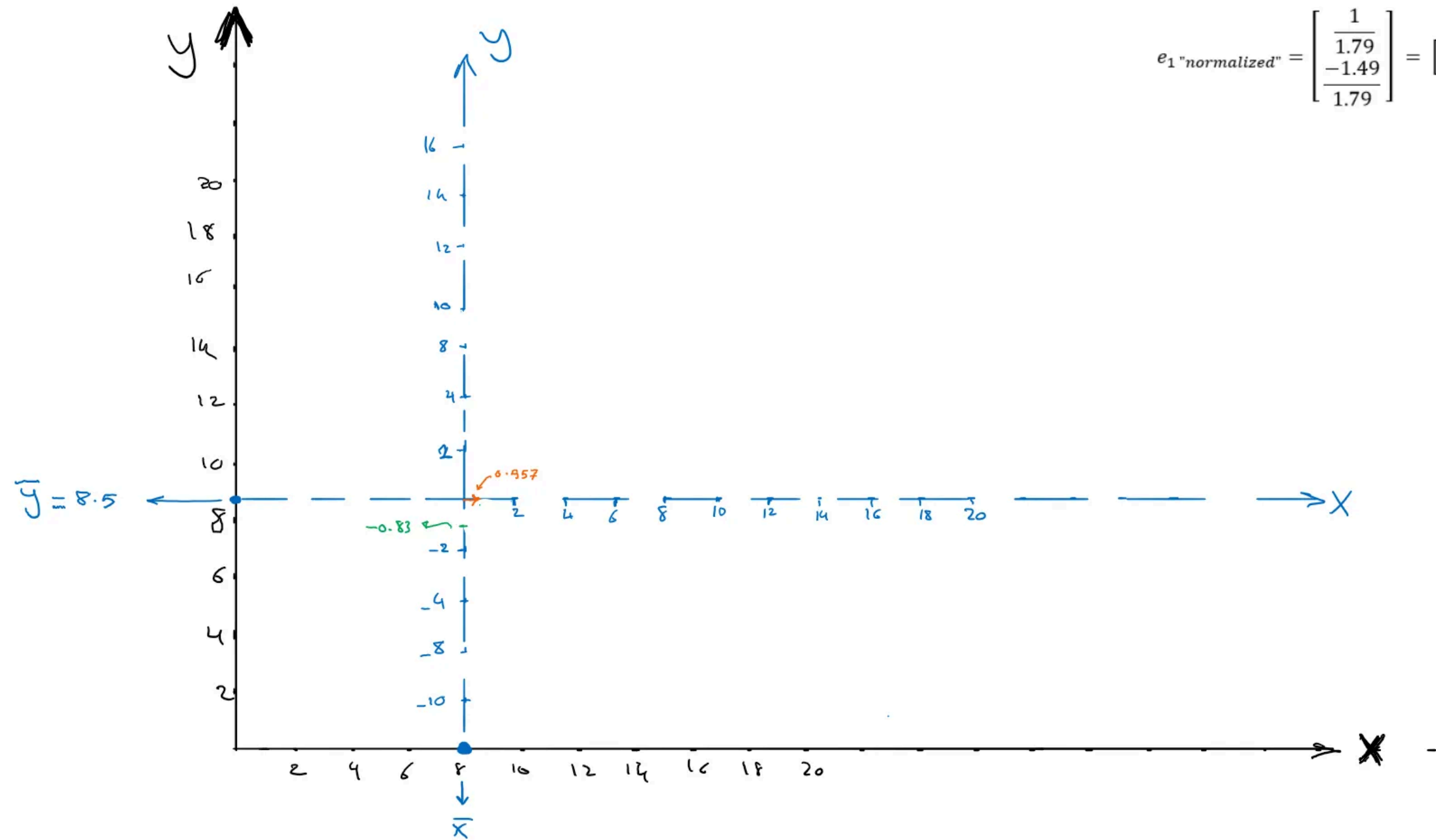


Figure (8)

- Plot the components of e_1 across the axes. (Note: the resultant vector slope can be calculated by dividing the 2nd component over the 1st component)



$$e_1 \text{ "normalized"} = \begin{bmatrix} \frac{1}{1.79} \\ \frac{-1.49}{1.79} \end{bmatrix} = \begin{bmatrix} 0.557 \\ -0.83 \end{bmatrix}$$

Figure (9)

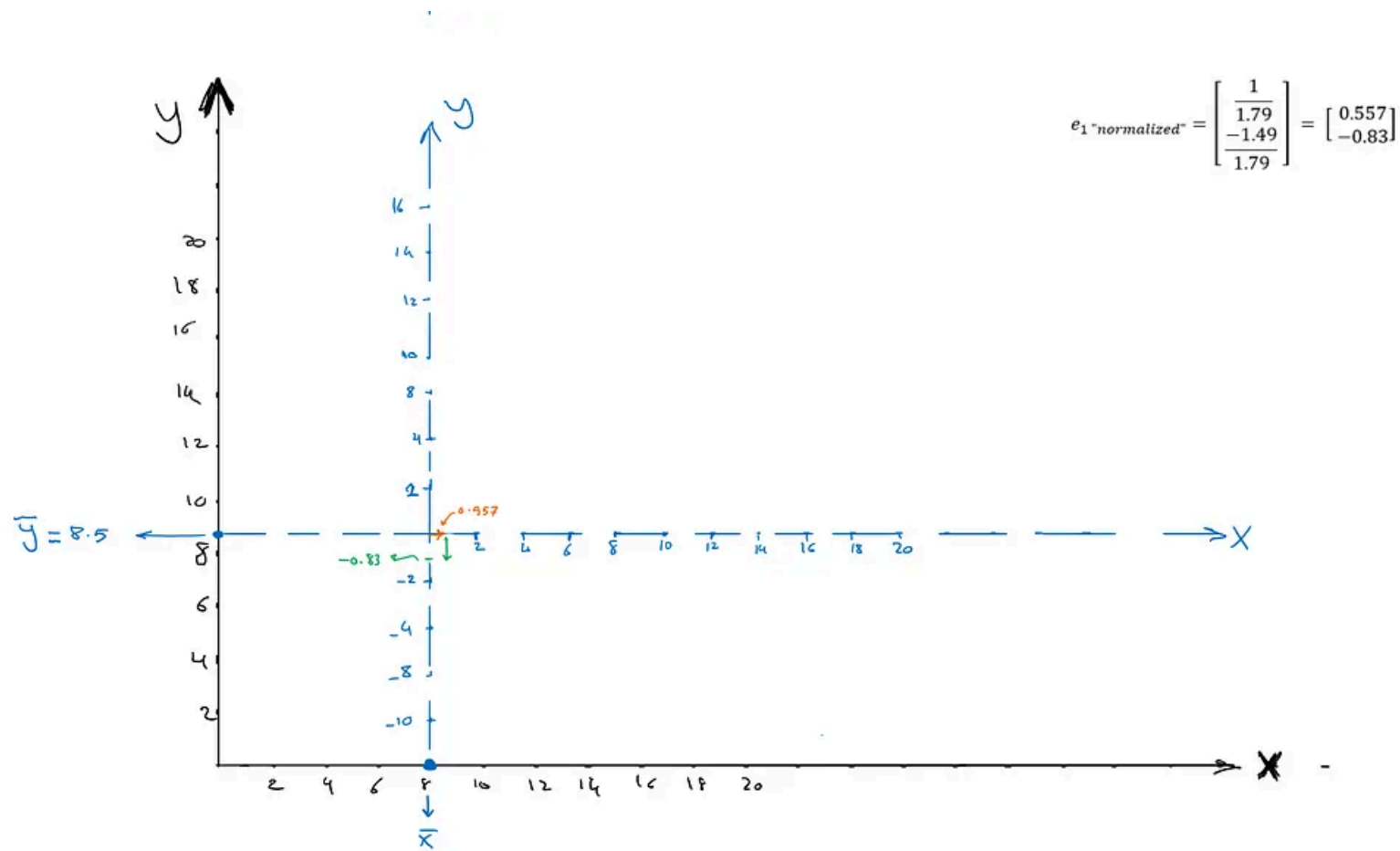
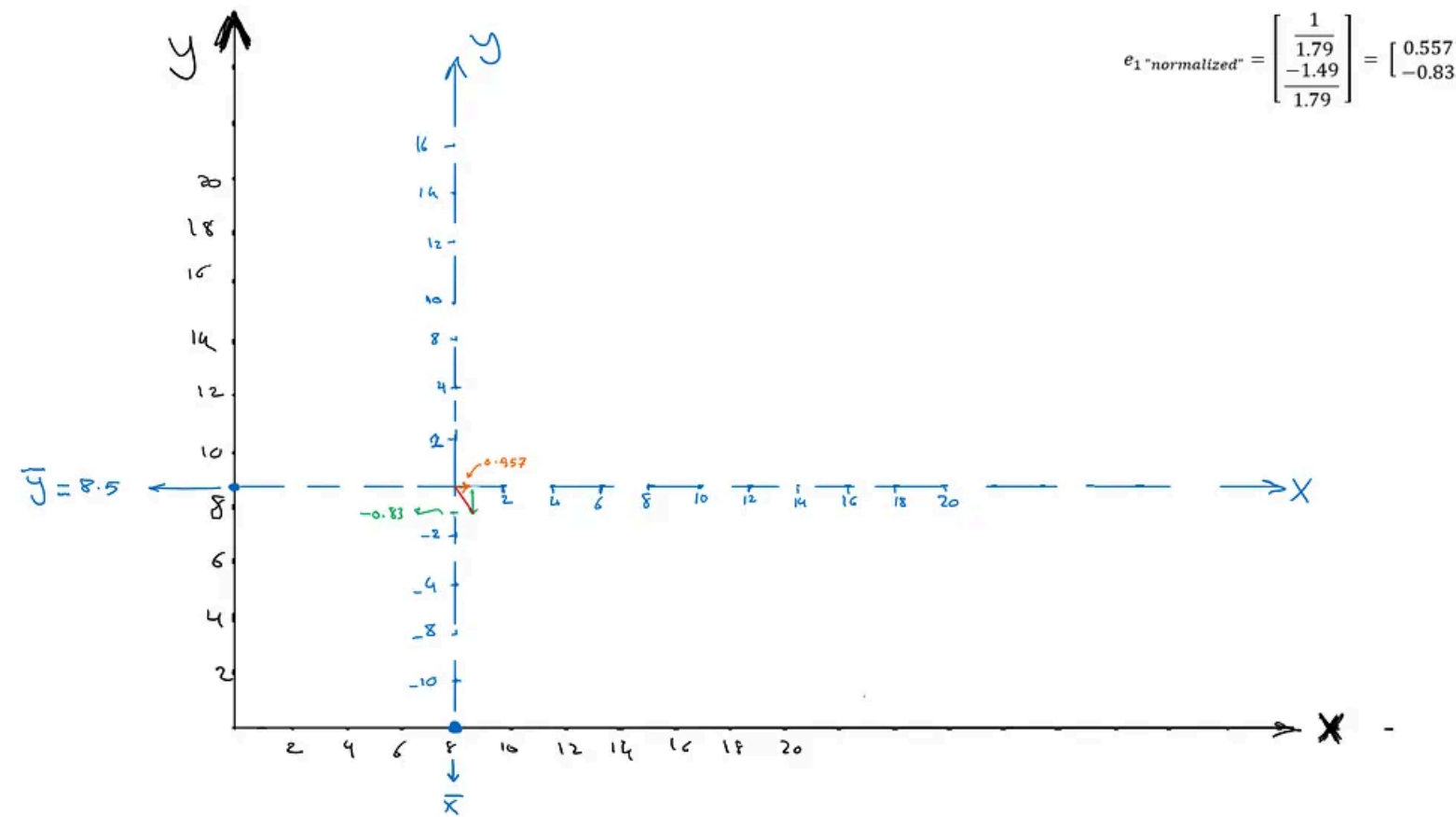


Figure (10)



$$e_1 \text{ "normalized"} = \begin{bmatrix} \frac{1}{1.79} \\ \frac{-1.49}{1.79} \end{bmatrix} = \begin{bmatrix} 0.557 \\ -0.83 \end{bmatrix}$$

Figure (11)

- Stretch the eigenvector e_1 axis slightly
- Do the same for e_2 . Notice how e_2 (PC2) is orthogonal to e_1 (PC1)

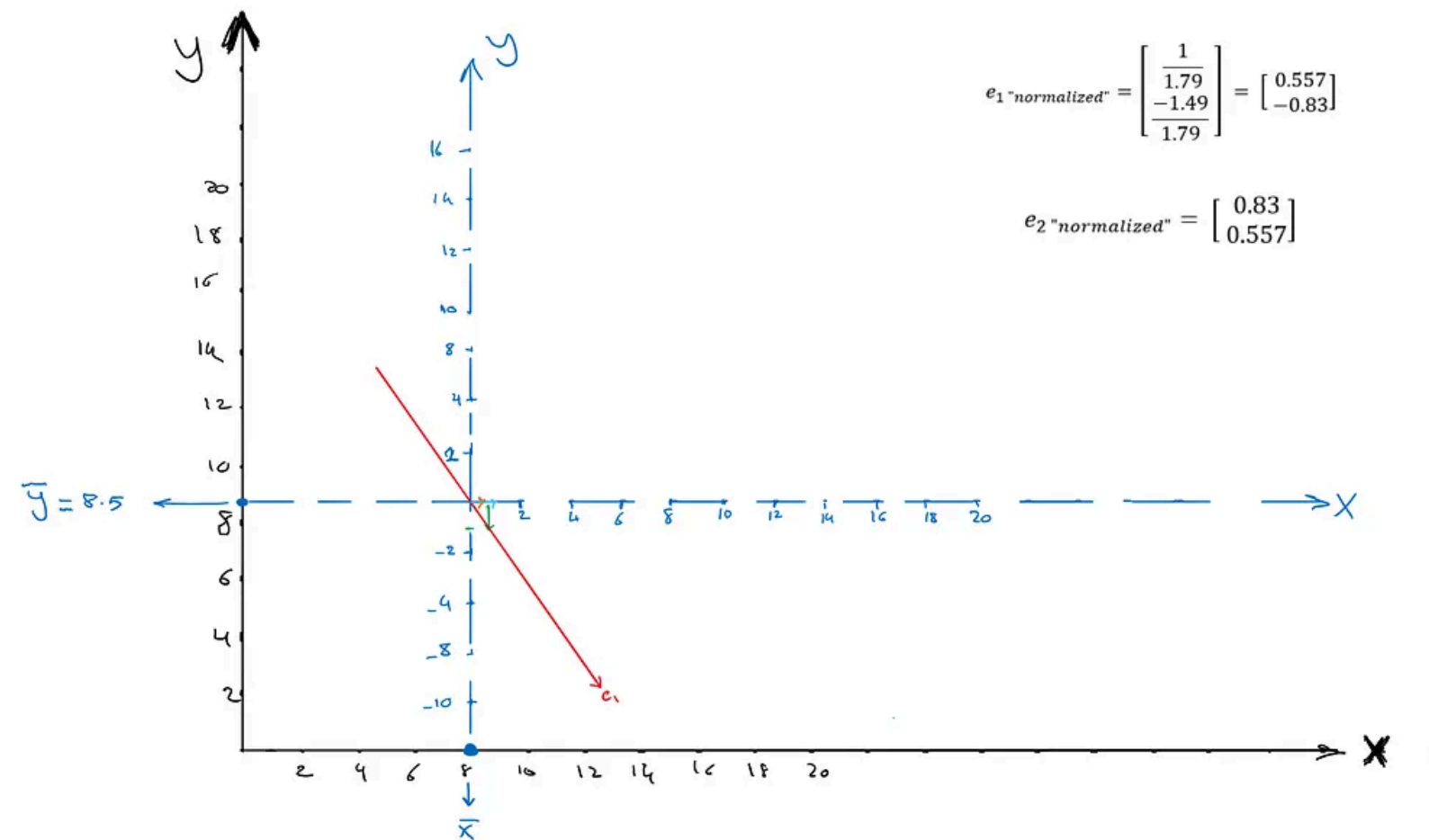


Figure (12)

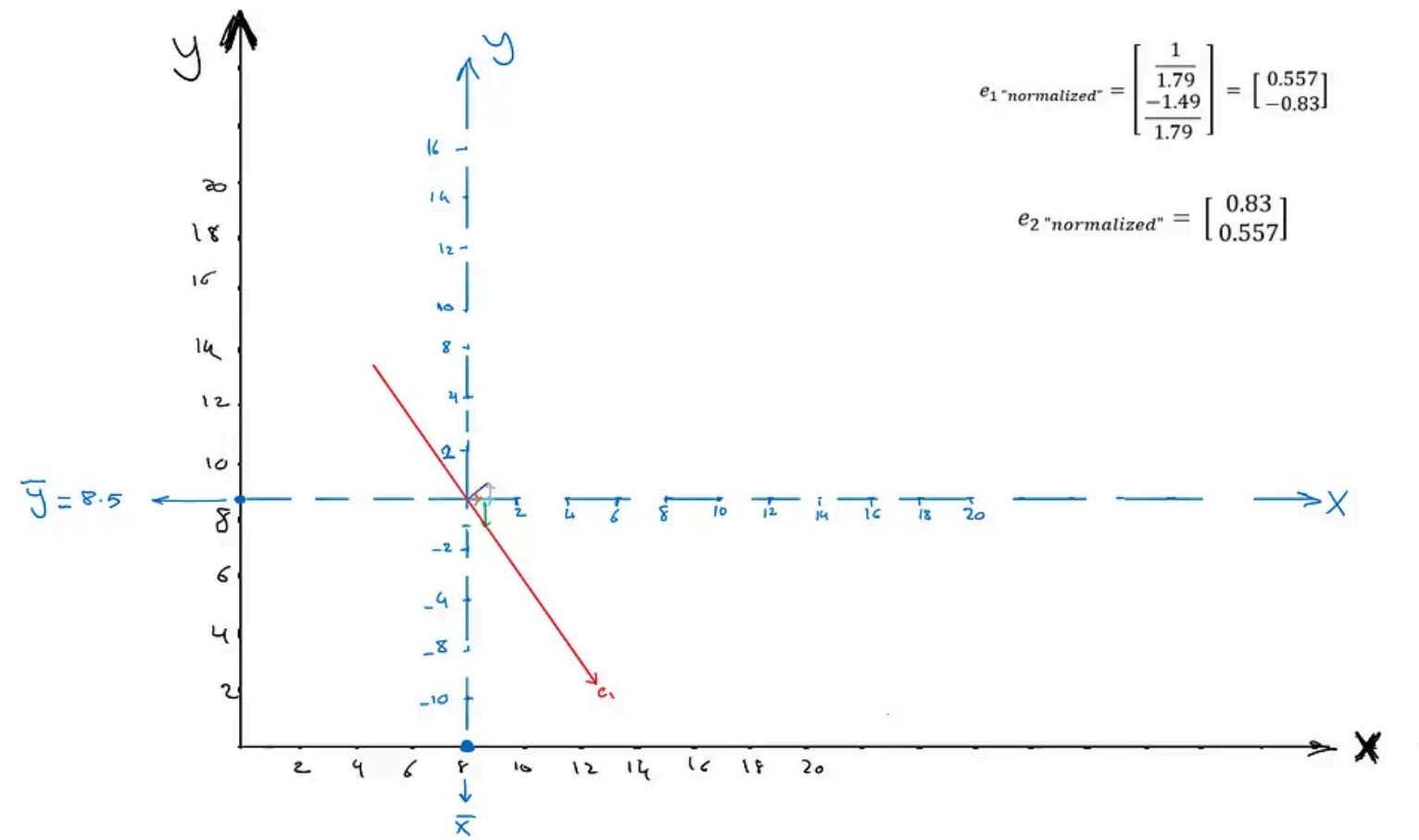


Figure (13)

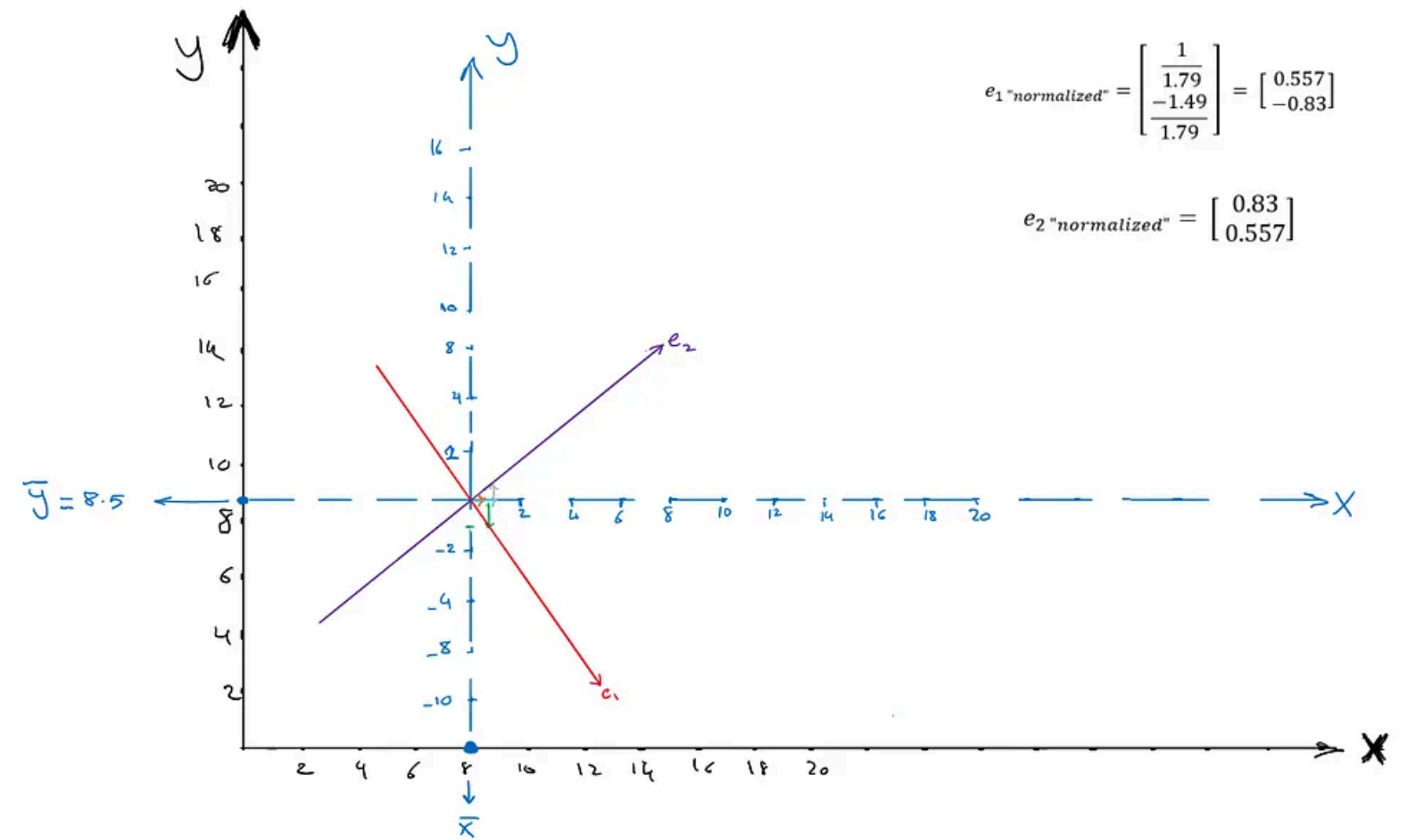


Figure (14)

- Using the new reduced dimension (PC1), plot the points from the table of the new data set:

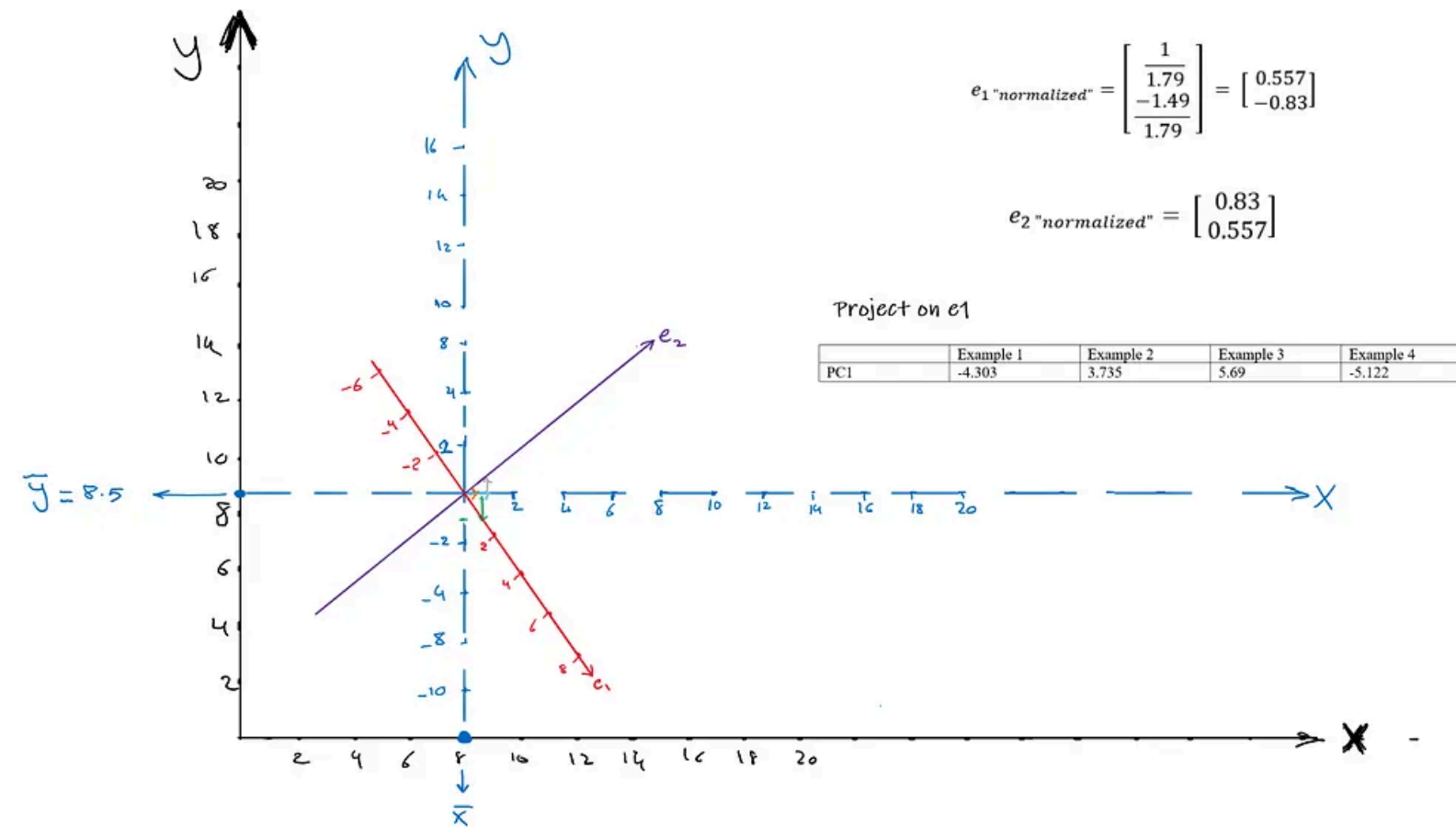


Figure (15)

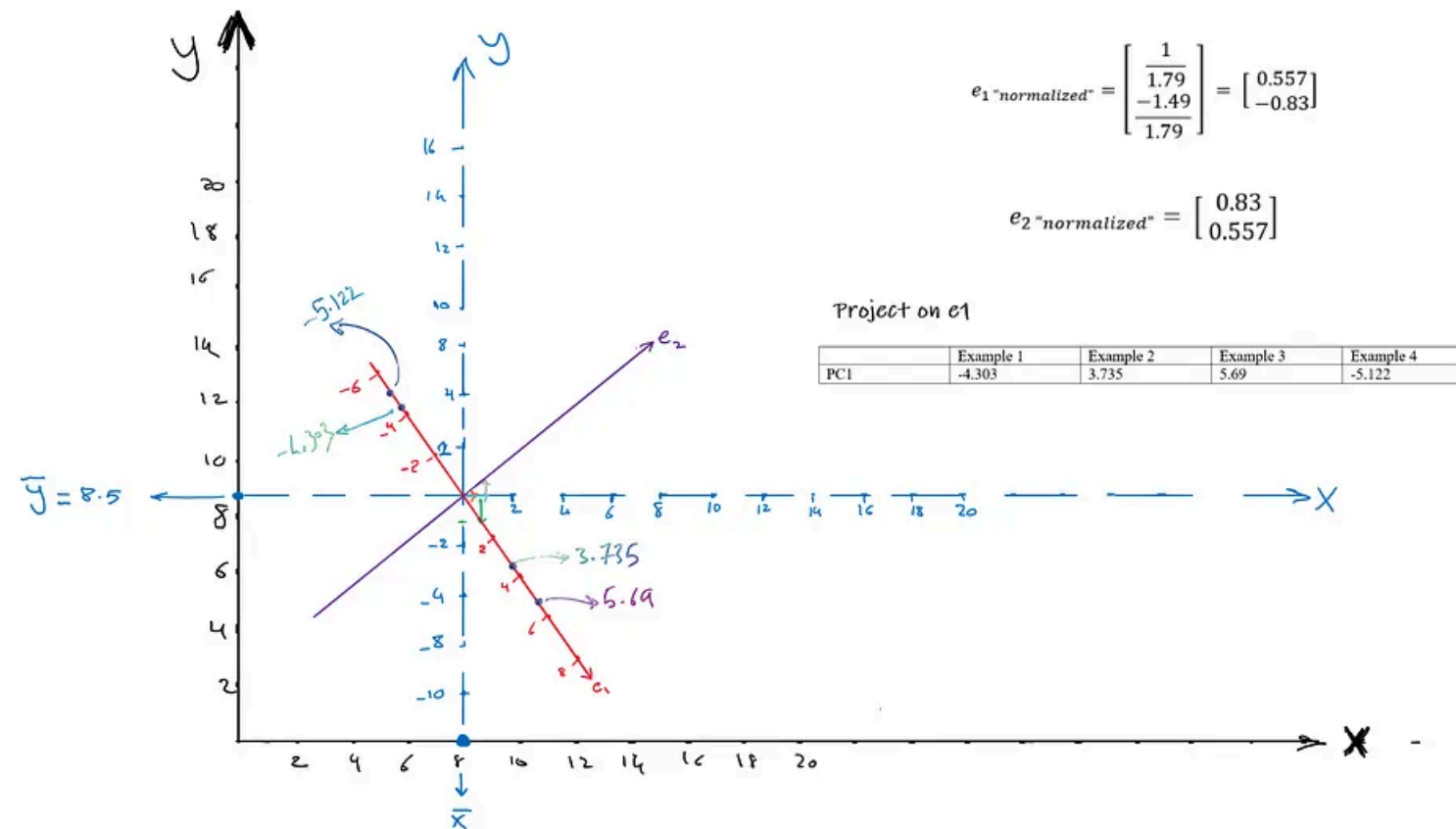


Figure (16)

Conclusion

We've went through each step of the PCA process in details, we solved for each one by hand, and we understood the goal of PCA, the match and linear algebraic notions behind it, when to use it and how many dimensions/principal components can we create and which dimensions do we use to fit our data.

Finally, if you want to watch the same tutorial as a video, you can check it on my YouTube channel (in Arabic):